Scalability & Future Plan

Date	28 October 2026
Team ID	EN25CS3010016
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Data is currently sourced from static CSV files.	Data updates require updating or reconnecting the source files.	High
2	Analysis is limited to the fields and records available in the supplied datasets.	New metrics and comparisons require additional data.	Medium
3	The current solution is primarily a Tableau workbook.	A standalone web interface and additional user access features would require further development.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Tableau workbook and dashboards.	Publish dashboards on Tableau Public or use a suitable hosted analytics solution for wider access.
2	Data Storage	Four CSV datasets connected directly to Tableau.	Use a centralized database or cloud data warehouse if the dataset grows significantly.
3	Performance	Visualizations operate on the connected datasets.	Optimize data preparation, relationships, and calculations as data volume increases.
4	Security	Access depends on how the workbook and published content are shared.	Apply appropriate access controls and avoid publishing sensitive or restricted data.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 1	Develop a Flask web application to present or embed the Tableau dashboards.	Short Term	Provides a more integrated project interface.
Phase 2	Add more Olympic datasets and historical performance metrics.	Medium Term	Enables broader comparisons and deeper analysis.
Phase 3	Automate data updates and preprocessing.	Medium Term	Reduces manual data maintenance.
Phase 4	Explore predictive analytics and advanced performance insights.	Long Term	May support forecasting and additional analytical capabilities.